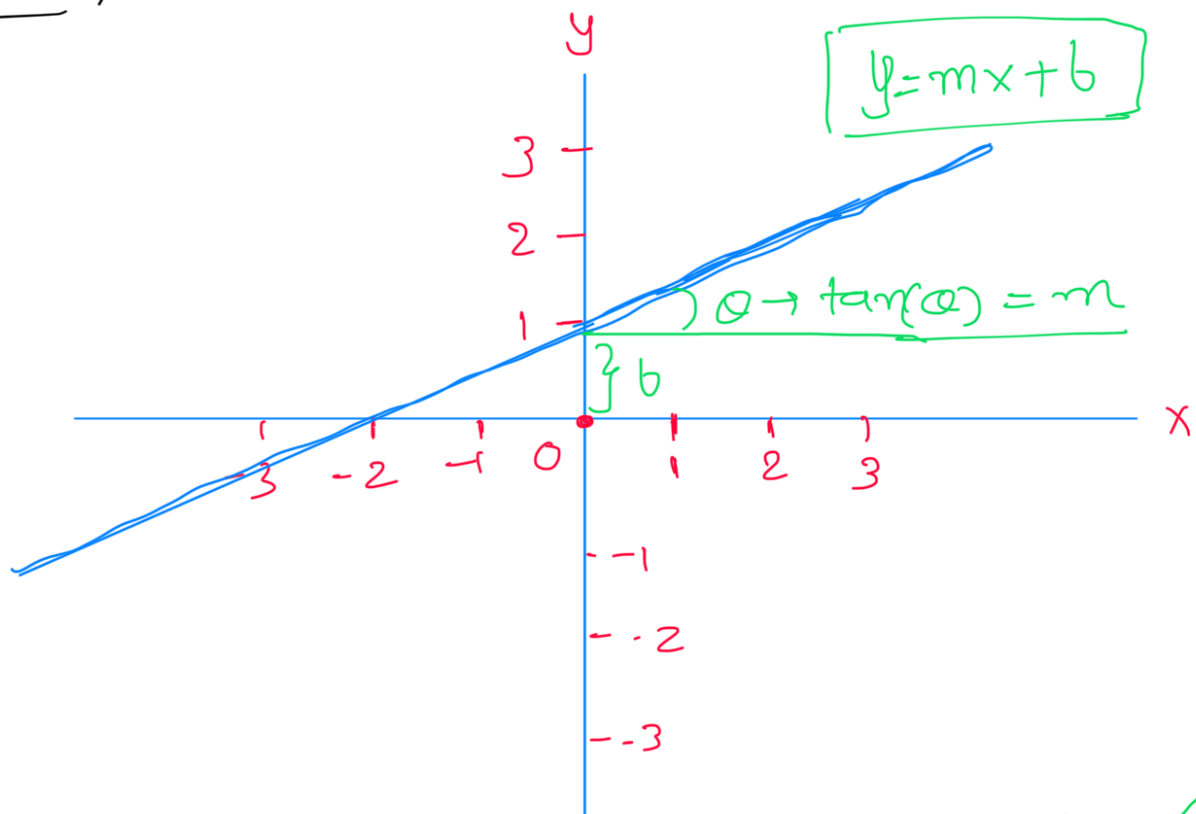


* geom ;



$y = mx + b \rightarrow$

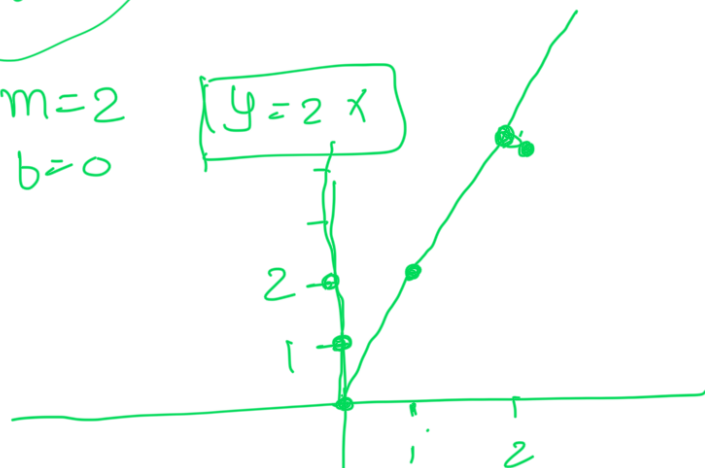
$m = 1$
 $b = 0$

$y = x$

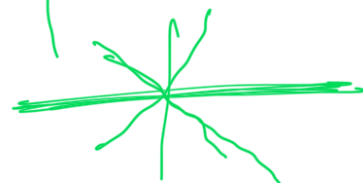


$m = 2$
 $b = 0$

$y = 2x$



$m = \infty$
 $b = 0$



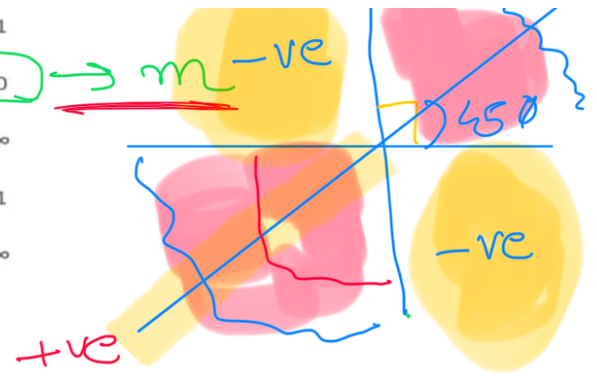
x	y
0	0
1	2
2	4

	0°	30°	45°	60°	90°	120°	180°	270°	360°
sin θ	0	1/2	1/√2	√3/2	1	√3/2	0	-1	0

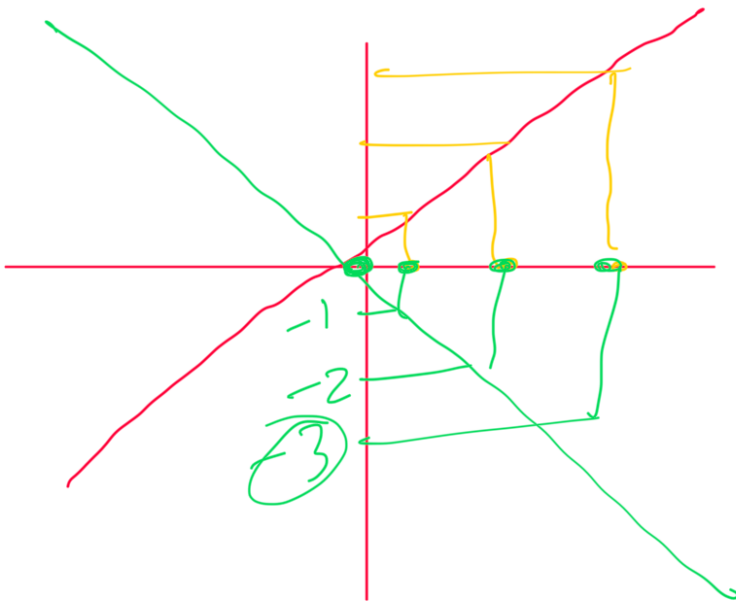
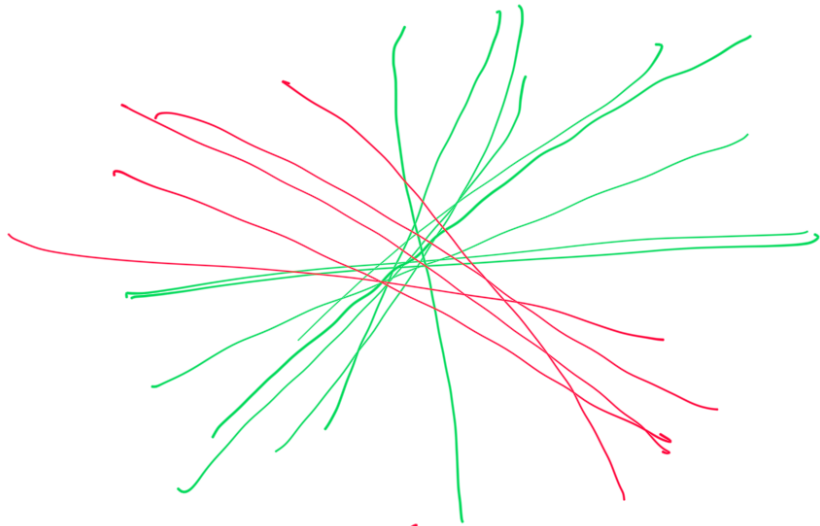
Trigonometric Table

+ve

$\cos \theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0	$-1/2$	-1	0	1
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞	$-\sqrt{3}$	0	∞	0
$\csc \theta$	∞	2	$\sqrt{2}$	$2/\sqrt{3}$	1	$2/\sqrt{3}$	∞	-1	∞
$\sec \theta$	1	$2/\sqrt{3}$	$\sqrt{2}$	2	∞	-2	-1	∞	1
$\cot \theta$	∞	$\sqrt{3}$	1	$1/\sqrt{3}$	0	$-1/\sqrt{3}$	∞	0	∞



$\nwarrow$
 +ve :
 -ve :



$$-3x - 1 \rightarrow 5$$

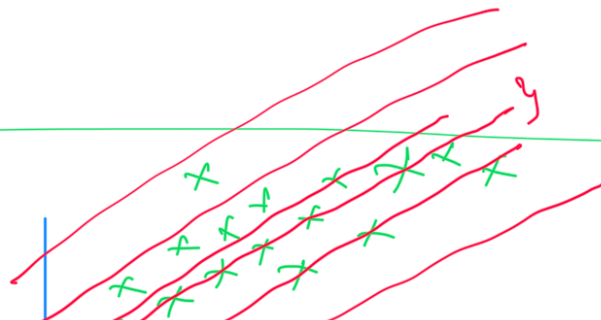
b = base value

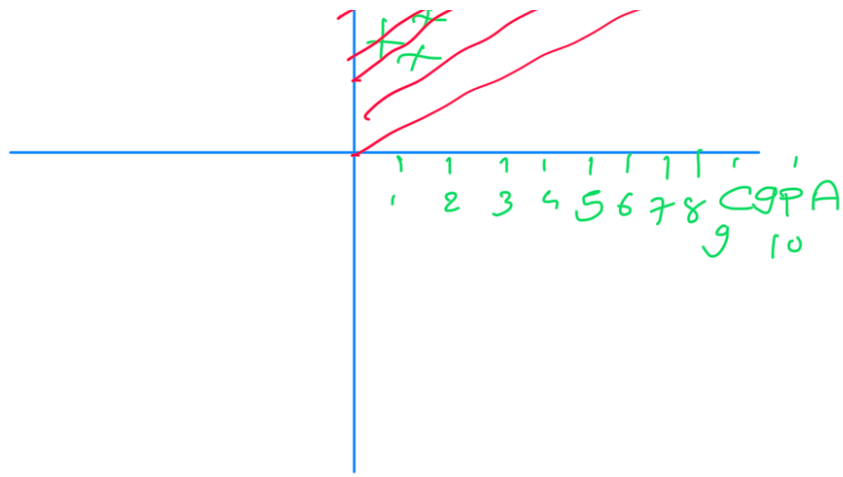
$y = mx + b$

$$\begin{aligned} b &= 1 \\ m &= 2 \end{aligned}$$

x y
 0 ①

LPA



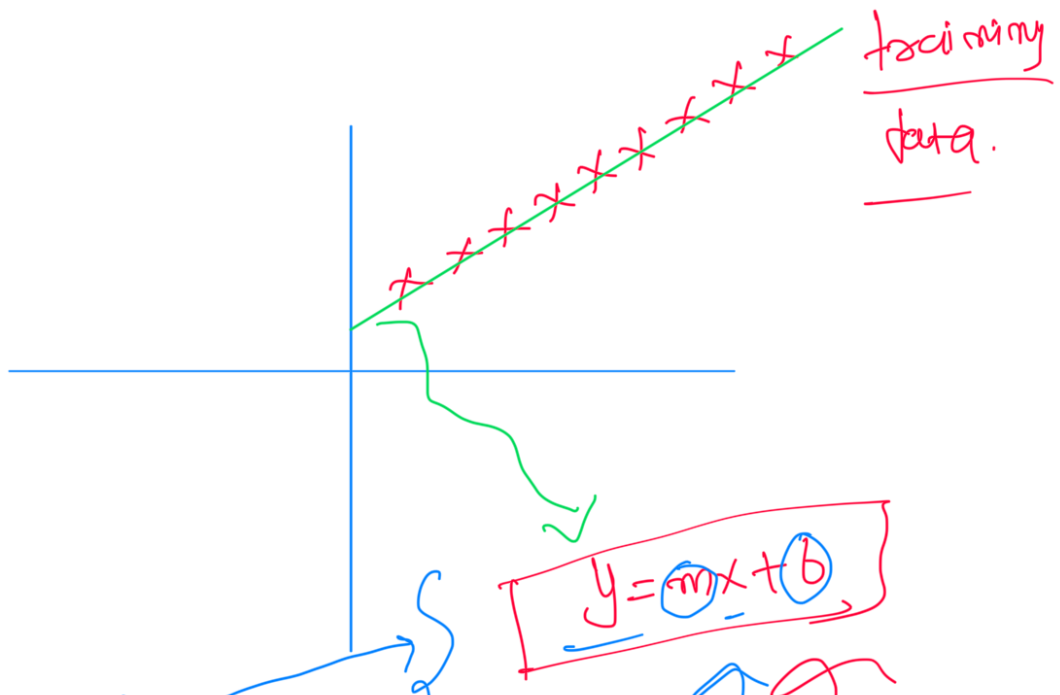
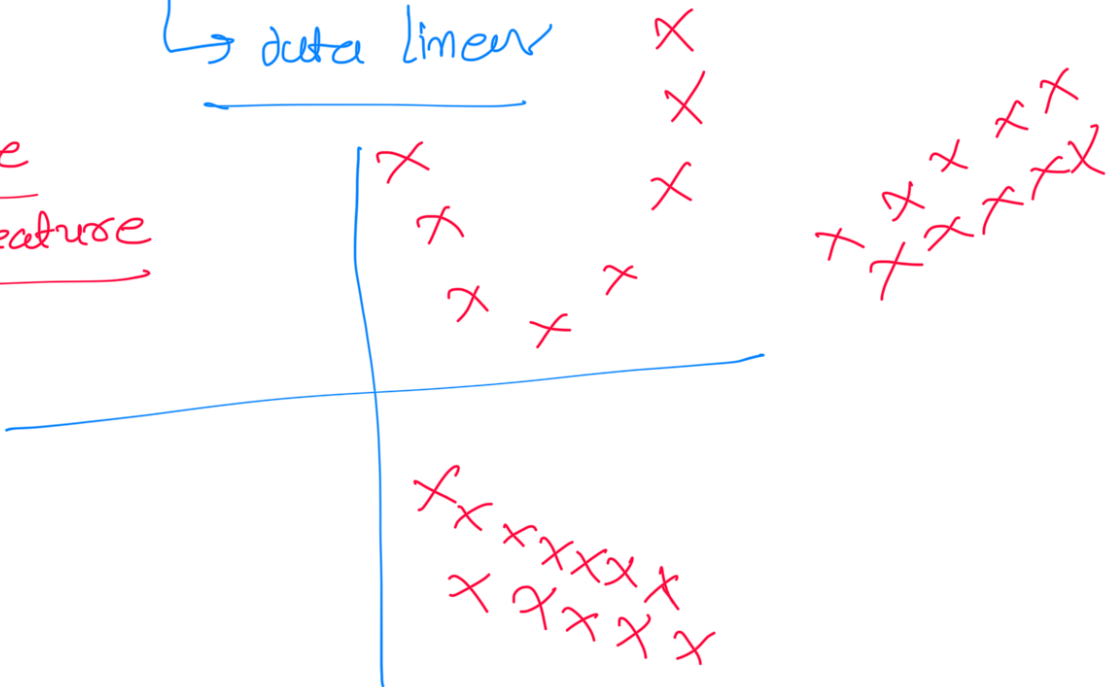


Simple Linear Reg. → numbers ✓



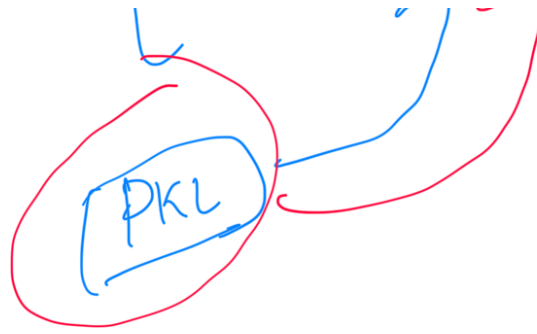
only one
I/p feature

↳ data linear



training -

How?



Q:

$$y = mx + b$$

$$\begin{aligned} m &= 4 \\ b &= 2 \end{aligned}$$

(y?)

$$x = 7.0$$

$$y = mx + b$$

$$= 4 \times 7 + 2$$

$$= 28 + 2 = 30$$